

Osmosis and Osmotic pressure of the solution

1. (c) $\pi = CRT = \frac{3 \times 1000}{180 \times 60} \times 0.0821 \times 288 = 6.56 \text{ atm.}$

2. (b) Isotonic solution $= \frac{w_1}{m_1 V_1} = \frac{w_2}{m_2 V_2}$

$$= \frac{w_1}{342 \times 1} = \frac{6}{60 \times 1} = \frac{342 \times 6}{60} = 34.2.$$

3. (c) $\pi = CRT$, $C = \frac{\pi}{RT} = \frac{0.0821}{0.821 \times 300} = 0.33 \times 10^{-2}.$

4. (a) $\pi = \frac{w}{m} \times RT = \frac{0.1}{1} \times 0.0821 \times 273$

5. (b) $\pi = \frac{n}{V} RT \Rightarrow M_P = \left(\frac{m}{V}\right) \frac{RT}{\pi}$

6. (b) $C = \frac{5}{342} \times \frac{1}{100} \times 1000 = \frac{50}{342} \text{ mol/l}$

$$\pi = \frac{50}{342} \times 0.082 \times 423 = 5.07 \text{ atm}$$

7. (c) $P = \frac{w}{mv} R.T$ since wvT are constant thus $P \propto \frac{1}{m}$

$$P_2 > P_1 > P_3.$$

8. (b) In the osmosis solvent molecule move from lower concentration to higher concentration.

10. (a) Osmosis occur from dilute solution to concentrate solution. Therefore solution A is less concentrated than B.

11. (c) Molar concentration of cane sugar $= \frac{5}{342} \times \frac{1000}{100} = \frac{50}{342}$

$$\text{Molar concentration of } X = \frac{1}{m} \times \frac{1000}{100} = \frac{10}{m}$$

$$\frac{10}{m} = \frac{50}{342} \text{ or } m = 68.4.$$



12. (d) Osmotic pressure method is especially suitable for the determination of molecular masses of macromolecules such as protein & polymer because for these substances the value of other colligative properties such as elevation in boiling point or depression in freezing point are too small to be measured on the other hand osmotic pressure of such substances are measurable.

13. (b) $\pi = CRT$; $C = \frac{\pi}{RT} = \frac{7.8}{0.082 \times 310} = 0.31 \text{ mol/litre}$

14. (b) $\pi = CRT$

$$\pi = \frac{w \times R \times T}{mV} = \frac{68.4 \times 0.0821 \times 273}{342} = 4.92 \text{ atm}$$

15. (a) **Normal saline solution**

Explanation:

Blood is isotonic with 0.9% NaCl solution, called normal saline solution. It has the same osmotic pressure as blood, so red blood cells neither shrink nor swell in it.

16. (b) $\pi = \frac{n}{V} RT = \frac{m/MRT}{V}$

$$\frac{600}{760} = \frac{20 \times 0.0821 \times 288 \times 1000}{500 \times M}; M = 1200$$

17. (c) $\pi = \frac{1.66 + 2.46}{2} = 2.06 \text{ atm}$

18. (a) **0.16 M NaCl**

Explanation:

Blood has the same osmotic pressure as approximately 0.16 M NaCl solution (about 0.9% saline solution). Therefore, it is isotonic with blood.

19. (d) Copper ferrocyanide *ppt.* acts as a semipermeable membrane.

20. (b) Osmotic pressure = CRT where $C = 1 \text{ m}$

$$\pi = CRT = 1 \times 0.0821 \times 300 = 24.6 \text{ atm}$$



